



Unit 4 · Outcome 3 · Scientific Investigation

The nature of evidence

KK06

VCE Environmental Science · Units 3 & 4 · Exam study summary

Key knowledge. 06 The nature of evidence Science doesn't run on belief, authority or how loudly someone argues

Digital Vector EnviroSci · single-topic mastery summary with worked good-vs-bad answers, exam traps, command words and practice.



When does data become evidence?

Foundations When does data become evidence? Students often use “data” and “evidence” as synonyms. They aren’t. The difference is the single most useful idea in this dot point. **D** Data are the raw recorded observations and measurements — Mara’s DO readings of 8.4, 8.1, 6.2 and 5.9 mg/L at four sites. On their own, numbers in a logbook prove nothing. **E** Evidence is data interpreted in relation to a specific claim. The moment Mara says “these falling DO values support my hypothesis that the drain lowers water quality,” the data has become evidence for some

Hypothesis, model & theory

The three things evidence is weighed against **Hypothesis, model & theory** The dot point names three different targets that evidence can support or refute. Examiners love this because students routinely muddle them — especially the idea that “theory” means “a hunch”. It’s the opposite. Click each tier in the model below. **3D** · live The certainty & scope ladder click a tier ↓ · **3D** view unavailable — use the tier buttons below; all content still works. **Hypothesis** Narrow · tentative · directly tested **Model** A simplified representation & predictive tool **Theory** Br

Does the evidence support or refute?

The verdict Does the evidence support or refute? Imagine a set of scales. On one pan, evidence consistent with the claim; on the other, evidence that contradicts it. The job of a scientist — and of you in the exam — is to read which way the beam tips, and how confidently. Add evidence to Mara’s scales and watch the verdict change. **3D** · interactive Mara’s scales of evidence add evidence ↓ **3D** balance unavailable — the verdict read-out below still updates. weigh each piece of evidence onto the scales: DO falls 8.4→5.9 mg/L downstream Pattern repeats on 3 s

Falsifiability & the asymmetry of proof

Why the two pans aren’t equal **Falsifiability & the asymmetry of proof** Here is the idea that separates an A+ answer from a B. Supporting and refuting evidence are not symmetrical. You can pile up a thousand confirmations and still not prove a hypothesis — but a single solid, reproducible contradiction can falsify it. A claim that could never be contradicted by any possible observation isn’t scientific at all. **3D** · animation Confirmations build a wall; one refutation tests it run the tests ↓ **3D** unavailable — the explanation and controls below still descri

Weight & quality of evidence

Not all evidence weighs the same **Weight & quality of evidence** A scientist doesn’t just count evidence — they weight it. Reproducible, peer-reviewed data from independent sources outweighs a single uncontrolled observation. The accumulated, converging weight of evidence is what builds scientific consensus — which is why “a study found” is much weaker than “decades of independent studies all find”. Strong evidence is... **Reproducible** Others repeating it get the same result. **Controlled** Variables managed so the effect is real, not a coincidence. **Independent** Con

Decode the command word

◆ COMMAND WORDS

Exam craft Decode the command word This dot point shows up under specific verbs. Each one wants a different shape of answer. Tap a verb to see what the examiner is really asking for.

Six exam traps

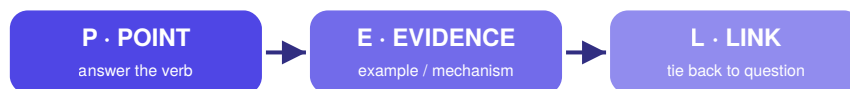
⚠ EXAM TRAPS

Lose-marks-here moments Six exam traps Every one of these is a real way students drop marks on “nature of evidence” questions. reveal the fix.

P·E·L trainer

🔗 P·E·L WRITING

Write like an examiner wants P·E·L trainer High-scoring evidence answers follow a rhythm: make a Point , cite the Evidence , then Link it back to whether the hypothesis is supported or refuted. Write a response below — the meters light up as you hit each move. Prompt Mara’s DO readings fall steadily from 8.4 mg/L upstream to 5.9 mg/L well downstream of the drain, and the pattern repeats across three sampling days. Evaluate whether this evidence supports her hypothesis that stormwater lowers water quality. 3 marks Point ○ A clear claim: does it support or



The P·E·L answer structure — make a Point, support with Evidence, then Link to the question.

Weak vs strong: a worked answer

✗ WEAK ANSWER

✗ Weak response — 1/4 A hypothesis is a guess about what will happen. A theory is also a guess, but a theory is a hypothesis that has been proven true . For example, climate change is a theory and DO going down is a hypothesis. Evidence proves both of them.

✓ STRONG ANSWER & MARKING

Why it loses marks: “theory = proven hypothesis” is the classic error; theories a



The P·E·L answer structure — make a Point, support with Evidence, then Link to the question.

Practice questions

★ PRACTICE & SELF-MARK

Test yourself · 21 marks Practice questions Attempt each in your logbook first, then reveal the marking guide and grade yourself honestly. Your score builds in the dock (bottom-right).

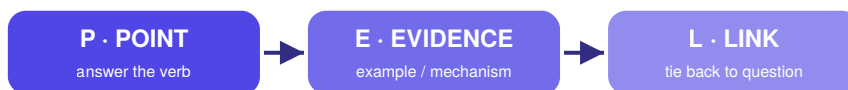
Recap — the seven things to remember

♦ RECAP / CHEAT SHEET

Lock it in Recap — the seven things to remember Data ≠ evidence Data become evidence only when interpreted in relation to a specific claim. Three different targets Hypothesis (narrow, testable), model (a representation/tool), theory (broad, well-supported explanation). Theory ≠ guess A theory is the strongest form of explanation, not a hypothesis awaiting promotion. Theories don't become "laws". Support, don't prove Evidence raises or lowers confidence . Science doesn't "prove"; use "supports". The asymmetry No amount of confirmation proves a claim, but

★ Good vs bad worked answers

Study the contrast — the same question answered two ways. The strong column shows the moves that earn the marks.



The P-E-L answer structure — make a Point, support with Evidence, then Link to the question.

✗ WEAK ANSWER

Unfalsifiable = unscientific. "The creek has a spirit that resists pollution" can't be contradicted by any measurement, so no evidence could ever bear on it. Outside science.

✓ STRONG / MODEL ANSWER

A hypothesis is a narrow, testable prediction — e.g. "DO is lower downstream of a stormwater drain" — which a single controlled investigation can support or refute . A theory is a broad, well-substantiated explanation supported by a large converging body of evidence — e.g. the enhanced greenhouse effect . A hypothesis relates to direct primary data from one study ; a theory rests on decades of independent, reproducible evidence and is the strongest form of explanation, not an unproven guess. Why it scores full marks: clear scope contrast (narrow vs broad) ✓ · correct example of each ✓ · the evidence distinction (one study vs converging body) ✓ · avoids "proof" language ✓ .